

REPRODUCIBILITY STATEMENT

Manuscript: Institutional Observability Under Scaled AI Governance: Deployment-Scale Capacity Degradation in Human Oversight Pipelines

Journal: Technology in Society (Elsevier)

Date: June 3, 2026

Reproducibility Commitment

This manuscript is accompanied by a complete reproducibility package that enables independent verification of all results.

What Is Reproducible

Component	Reproducible?	Method
State-space model derivations	Yes	Mathematical proofs in manuscript + supplementary appendix
Kingman heavy-traffic analysis	Yes	Closed-form derivation (Eqs. 3–4)
Monte Carlo simulation	Yes	Python code in GitHub repository
SEDI index computation	Yes	Python code in GitHub repository
Figure 1 (saturation curve)	Yes	Generated from simulation code
Figure 2 (divergence plot)	Yes	Generated from simulation code
Figure 3 (routing architecture)	Yes	Conceptual diagram; source files provided
Figure 4 (review depth KDE)	Yes	Generated from simulation code
Figure 5 (SEDI degradation)	Yes	Generated from simulation code
Table 1 (parameters)	Yes	Documented in manuscript and code

What Is Not Reproducible

Component	Reason
Empirical validation against real institutional data	No primary data collected; theoretical study
Calibration data	From published public sources (Vanderbilt, PHTI); not independently collected
Reviewer behavior measurements	Parameterized from literature, not directly measured

Reproduction Instructions

Environment

- Python 3.10+
- Required packages: `numpy`, `scipy`, `matplotlib`, `pandas`, `seaborn`
- Install: `pip install -r requirements.txt`

Steps to Reproduce

1. Clone repository: `git clone [repository URL]`
2. Install dependencies: `pip install -r requirements.txt`
3. Run Monte Carlo simulation: `python monte_carlo_simulation.py`
4. Generate figures: `python figure_generation.py`
5. Compute SEDI: `python sedi_computation.py`

Expected Runtime

- Monte Carlo simulation (120 runs): ~30 seconds on standard hardware
- Figure generation: ~10 seconds
- Total reproduction time: <5 minutes

Expected Outputs

- CSV files with simulation outputs (queue lengths, review depths, SEDI values)
- PDF figures matching those in the manuscript
- Console output with key statistics

Repository

GitHub: <https://github.com/SilverCoin256/oversight-saturation-sedi> **Zenodo DOI:** [DOI to be minted upon acceptance] **License:** MIT

Statement prepared by: Shaurya Gupta **Date:** June 3, 2026